Nazib Abrar

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Education

Rajshahi University of Engineering & Technology 2022 - 2026
BSc in Mechatronics Engineering | CGPA 3.23/4.00

Notre Dame College, Dhaka 2020
HSC in Science | GPA 5.00/5.00

Work Experience

Hardware and Systems Engineer | Siliconova February 2026 - Present

- Assisted in SQA and testing of a high-speed networking application, exercising DPDK kernel-bypass networking and SPDK kernel-bypass management of high-speed SSD operations.
- Developed licensing and gRPC control applications and a Qt 6 GUI for a multi-device system integrating SDRs, FPGAs, and high-performance servers for satellite data acquisition and recording.
- Managed the technical documentation and packaging of client deliverables.

Embedded Systems Engineer (Contract) | Aethes Armor (USA) March 2025 - January 2026

- Engineered a Debian based custom embedded Linux distribution for a security locker device using the Yocto Project, including writing custom device drivers and patching the kernel.
- Developed the device's core on-device application stack, implementing WebRTC for live video-calling and WebSockets for real-time control of peripherals.
- Integrated computer vision and AI functionalities using TensorFlow-Lite to enable intelligent device operations.
- Built the complete web backend and server infrastructure for the system using Django.
- Contributed to the hardware development lifecycle by assisting in PCB design reviews and Bill of Materials (BOM) generation.

Junior Software Developer (Part-Time) | FronTech Limited June 2023 - October 2025

- Created well-documented libraries in C++ for JRC Board (an ESP32-based microcontroller development board) to ensure compatibility with Arduino shields.
- Led the back-end development for the "smartschoolbus.gov.bd" project, a B2B solution built with NodeJS, Express, and MongoDB to improve school bus safety and efficiency for over 2,500 active users.
- Contributed to the firmware development of the IoT attendance and vehicle tracking device deployed on the buses.
- Developed a strategic algorithm that resolved network-instability-induced gaps in the device's attendance and tracking records through an offline synchronization mechanism.
- Managed VPS infrastructure, utilizing Nginx for traffic routing and access control, and implemented CRON jobs for automated database backups.
- Implemented live video streaming in RTMP and HLS protocol.
- Integrated RabbitMQ and Celery to handle asynchronous tasks, enabling efficient processing of time-consuming operations, and deployed services in a microservice architecture using Docker for scalability and seamless deployment.

Research & Projects

StereoLite: A Novel Stereo Vision Architecture for Disparity Matching and Depth Estimation

- Designed a novel lightweight stereo matching architecture that fuses a tile-hypothesis cost-volume paradigm with an iterative refinement loop and learned convex upsampling, targeting real-time depth estimation on edge hardware.
- Conducted a systematic literature review of $\sim$190 deep stereo matching papers and authored a comparative analysis spanning classical SGM to foundation-model-era methods, informing the architecture's design choices.
- Ran a multi-phase ablation study (9 architectural variants A/B-tested in parallel on Modal A100s) covering cost-volume designs, slope-aware warping, context branches, and several loss-side interventions to isolate the contributing components.
- Implemented the model in PyTorch with a custom multi-scale tile refinement loop and groupwise 1D cost volume, benchmarked against published lightweight stereo baselines on Scene Flow and indoor datasets.

CORTEX-Health: An AI Assistant for Medical Practitioners

- Developed a FastAPI-based API server to facilitate communication between an Android application (built with Flutter) and machine learning models. Used PostgreSQL as the database.
- Trained three YOLO-v8 models using PyTorch for disease diagnosis from medical images (e.g., X-RAY and CT scan reports).

Vision-Based Autonomous Drone Navigation in GPS-Denied Environments

- Researched and prototyped an onboard perception and navigation stack for a PX4-based drone to fly autonomously in GPS-denied environments, fusing monocular and lightweight stereo vision for visual odometry and depth estimation.
- Integrated MiDaS monocular depth estimation with YOLOv8 semantic segmentation for environmental understanding, and implemented an A*-based local planner for collision-avoidance trajectory generation.
- Patched the PX4 flight controller firmware to disable GPS requirements and bridged the on-Raspberry Pi vision stack to PX4 over MAVLink, enabling closed-loop steering from onboard perception to motor commands.
- Built a parallel ROS 2 Jazzy + Gazebo Harmonic simulation environment with per-pixel labeled worlds and a Blender-based mesh pipeline to validate the perception, planning, and control loop end-to-end before flight tests.

Leadership and Awards

Champion - Bangladesh Round, 4th Global - 6th Kibo RPC 2025

- Achieved 1st position in the Bangladesh Round and 4th globally in the Kibo RPC, organized by JAXA in cooperation with NASA, developing software to control NASA's Astrobee free-flying robot aboard the "Kibo" Japanese Experiment Module of the International Space Station. Employed computer vision techniques for navigation and task execution in zero gravity.

Global 11th Position – International Rover Design Challenge 2023

- Contributed to Team Ogrodoot's best performance on IRDC by implementing computer vision based mars terrain segmentation system using SAM and designing inverse kinematics algorithm for robotic arm of the rover.

First Runner-Up - Phitron Show Your Project Contest 2022

- Won the project showcase competition with CORTEX Robotic Arm Controller Software.

Champion - DRMC Tech Carnival 2020

- Won the Line Follower Robot racing competition using "Thunder," an LFR controller software developed with C++.

Finalist - Code Samurai 2024 (Hackathon)

- Advanced to the final round in a competitive field of 400+ teams, achieving 14th place in the Code Samurai 2024 Hackathon with an end-to-end solution for municipal solid waste management.

Secretary of Programming Dept | Notre Dame Information Technology Club

- Led programming-related affairs, including contests and tutoring sessions for junior members.

Problem-Solving Experience

- Solved 85+ problems on LeetCode following the Neetcode.io roadmap, gaining a strong understanding of diverse data structures and algorithms. [Profile Link]
- Codechef Rating 1550. [Profile Link]

Certificates

Machine Learning Specialization September 2023
Issued by: Stanford Online

Attachment

Projects, Platforms, and Verification

The following pages provide further detail on the work experience, platforms, and projects referenced in the preceding résumé, together with demonstrations of the delivered systems and verification of the listed credentials.

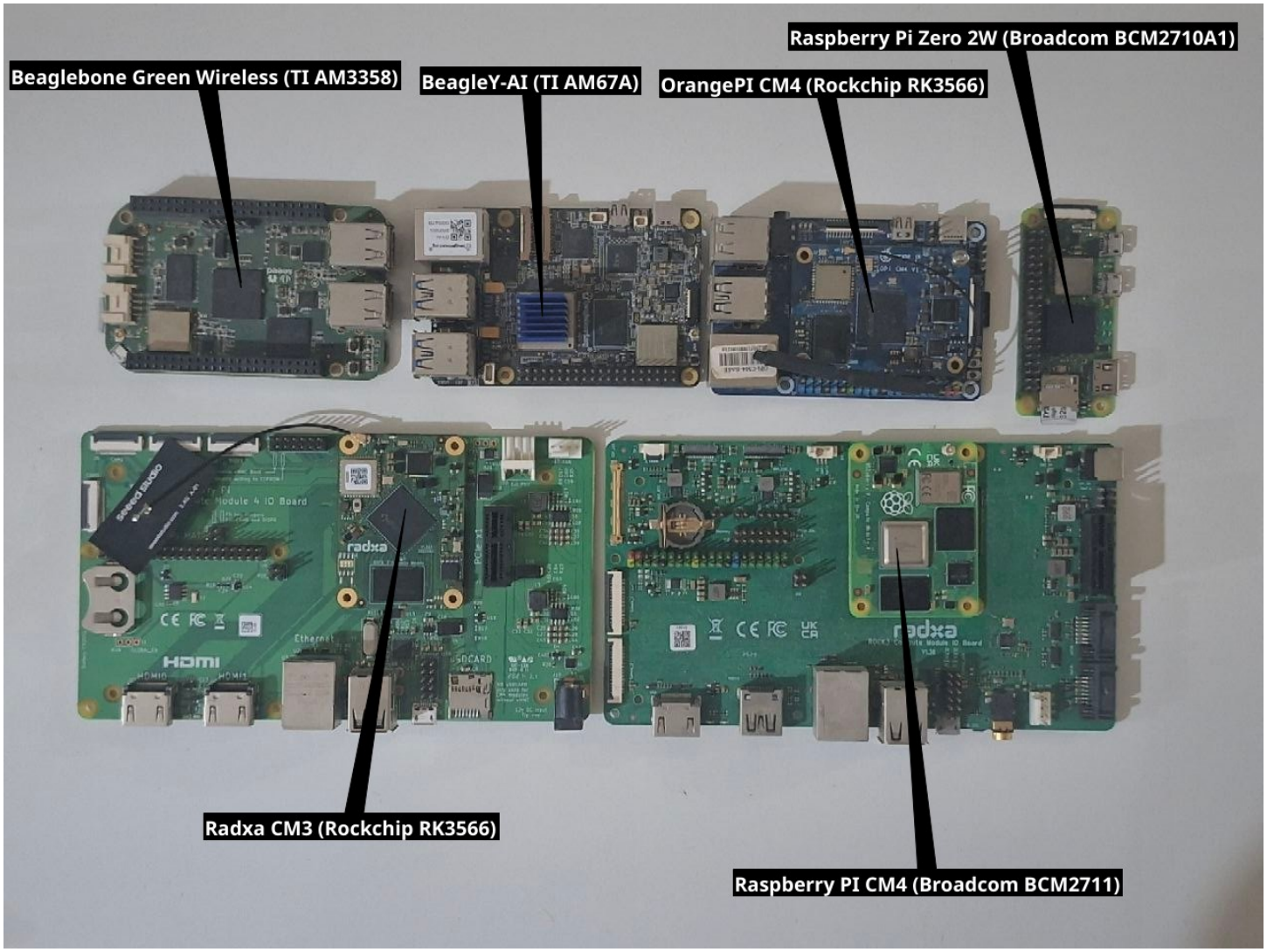
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Technical Attachment: Projects, Platforms, and Verification

Embedded Linux Platforms

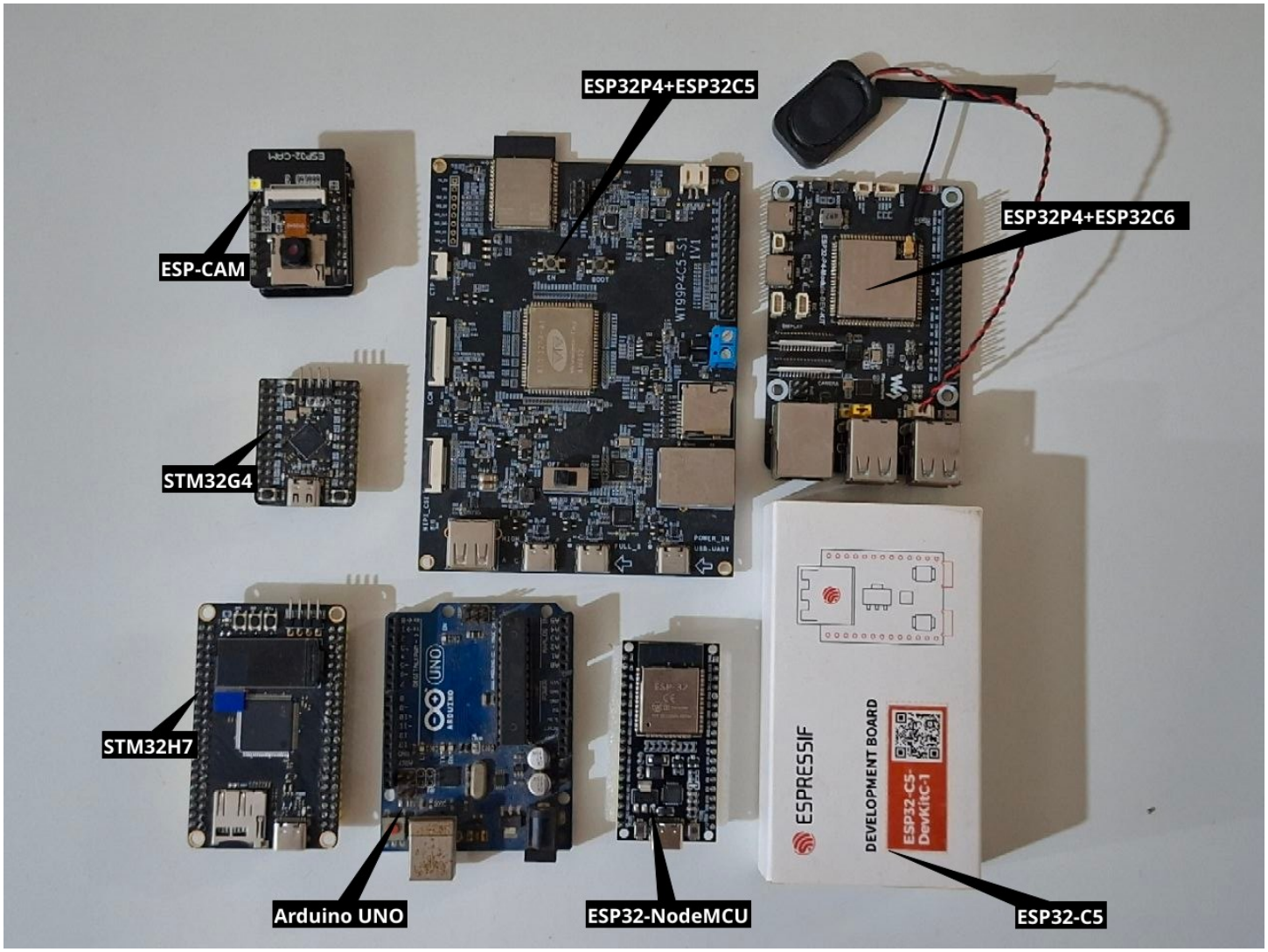
- **OrangePi CM4 and Radxa CM3** (Rockchip RK3566). Client project comprising custom carrier board development, an RKNN based computer vision stack, and a hardware accelerated GStreamer pipeline, deployed on a custom Yocto distribution built for both boards.
- **Raspberry Pi CM4 and Raspberry Pi Zero 2W** (Broadcom BCM2711 and BCM2710A1). Client projects covering a security delivery locker with WebRTC video conferencing and hardware accelerated video recording, including GStreamer pipeline development using `v4l2src`.
- **BeagleY-AI and BeagleBone Green Wireless** (Texas Instruments AM67A and AM3358). Conducted tutoring sessions on the Texas Instruments SDK at the RUET IoT Club, covering device driver development and the AI ISP and NPU hardware acceleration SDKs.



Single board computers and compute modules used in the projects described above.

Microcontroller Platforms

- **STM32H7 and STM32G4**. Client project comprising firmware development for a CAN analyzer capable of inspecting a live CAN bus and streaming its traffic for real time data visualization.
- **ESP32-P4 with ESP32-C5, and ESP32-P4 with ESP32-C6**. Client project implementing non-AI video calling over WebRTC using `esp-peer`.
- **ESP32-CAM**. Employed as the camera node across a range of projects.
- **Interfaces and toolchains**. Practical experience with the SPI, I2C, and CSI camera interfaces, and firmware development in STM32CubeIDE, STM32CubeMX, ESP-IDF, and the Arduino IDE.



Microcontroller development boards used for firmware and peripheral driver work.

Project Demonstrations

SmartSchoolBus (smartschoolbus.gov.bd), a Government of Bangladesh School Bus Safety Platform

- National television coverage of the deployed product by Somoy TV: youtu.be/vm3ZXsQ_LZ8
- Operational walkthrough of the platform, presented in the capacity of lead engineer on the project: youtu.be/xTgJycR2Adk

Vision-Based Autonomous Drone Navigation in GPS-Denied Environments

- Research walkthrough covering the approach, perception stack, and planning: youtu.be/0HMXJcbxA3w
- Flight demonstration: youtu.be/mjQc9S_NIv8

Robotic Arm Inverse Kinematics Control

- Inverse kinematics with visual feedback, servo driven: youtu.be/yXRN9Aov4zM
- Inverse kinematics, stepper driven: youtube.com/shorts/-wWf4OxTI5E

Line Follower Robot "Thunder" (Champion, DRMC Tech Carnival 2020)

- Short demonstration of the C++ controller software: youtube.com/shorts/ZHi7nnWHWrQ

Kibo Robot Programming Challenge (JAXA and NASA)

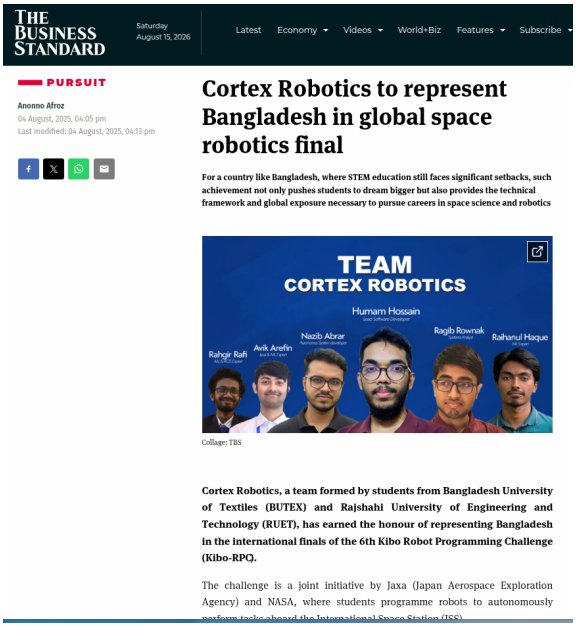
- The Kibo-RPC is hosted by the **Japan Aerospace Exploration Agency (JAXA)** in cooperation with **NASA**. Participating teams develop software that autonomously controls **Astrobee**, NASA's free-flying robot, inside the "Kibo" Japanese Experiment Module aboard the **International Space Station**.
- Competing as **Team Cortex Robotics** (Bangladesh), the team secured first position in the national round and **fourth position globally** in the on-orbit final, completing the mission in 3:42 with three of four correct image matches and correct identification of the target item.
- **On-orbit execution of the team's program aboard the ISS:** youtu.be/sHPJ4LExIgM

JAXA Human Spaceflight Technology Directorate Humans in Space

For Corporate & Research Institute Experiment at Kibo

Rank	Country/Region	Team Name	Mission Completion Time	Number of Correct Matches	Found th correct ite
1	Taiwan	iTron	3:34	4/4	OK
2	Malaysia	Automen	2:45	4/4	NG
3	Philippines	Inflection Point	4:21	4/4	NG
4	Bangladesh	Cortex Robotics	3:42	3/4	OK
5	Indonesia	Narantaka GMAT	3:11	2/4	OK
6	Japan	FUNABASHI	2:42	3/4	NG
7	USA	StarStrikers	Time Over	1/4	NG

Official JAXA final round standings.
Rank 4, Bangladesh, Cortex Robotics.



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PURSUIT

Anonno Afroz
04 August, 2025, 04:05 pm
Last modified: 04 August, 2025, 04:13 pm

Cortex Robotics to represent Bangladesh in global space robotics final

For a country like Bangladesh, where STEM education still faces significant setbacks, such achievement not only pushes students to dream bigger but also provides the technical framework and global exposure necessary to pursue careers in space science and robotics

Collage: TBS

Cortex Robotics, a team formed by students from Bangladesh University of Textiles (BUTEX) and Rajshahi University of Engineering and Technology (RUET), has earned the honour of representing Bangladesh in the international finals of the 6th Kibo Robot Programming Challenge (Kibo-RPC).

The challenge is a joint initiative by Jaxa (Japan Aerospace Exploration Agency) and NASA, where students programme robots to autonomously perform tasks aboard the International Space Station (ISS)

The Business Standard, 4 August 2025.
National coverage of the team's qualification.

Verification and Certificates

- **International Rover Design Challenge 2023**. Eleventh position globally with Team Ogrodoot (RUET), issued by the Space Robotics Society.
- **Machine Learning Specialization**. Stanford Online and DeepLearning.AI, September 2023. Verification: coursera.org/verify/specialization/T6GAYS73CU62



International Rover Design Challenge 2023, Space Robotics Society.



Machine Learning Specialization, Stanford Online and DeepLearning.AI.